

Notes: Baker (JPE, 1992)

Incentive Contracts and Performance Measurement

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1 Big picture

- **Question.** How should an organization design incentive contracts when the principal's true objective is not contractible, and the only available contractible variables are imperfect performance measures?
- **Core departure from standard agency theory.** Standard moral hazard models often assume that the principal can contract on the true output or value objective, perhaps with additional signals. Baker starts from the opposite problem: the principal's objective itself may be unavailable as a performance measure.
- **Key examples.** Nonprofits, government agencies, partnerships, private firms, and many divisions inside public firms often cannot contract on total organizational value. They use proxies such as patents, sales revenue, accounting profit, divisional targets, or relative performance.
- **Central mechanism.** The agent observes state-specific information before choosing effort. The performance measure tells the agent what to do only imperfectly. If the performance measure's marginal response to effort differs from the true value marginal response, incentives are distorted.
- **Main result.** The optimal linear piece rate is lower when the performance measure's marginal product is poorly aligned with the value marginal product. This dampening occurs even when the agent is risk neutral.
- **Metric for a good performance measure.** A performance measure is good when P_e and V_e are highly correlated, where P_e is the marginal product of effort on the performance measure and V_e is the marginal product of effort on true value.
- **First-best condition.** With unobservable effort, the contract is first-best only if the performance measure responds to effort exactly as value does in every state: $P_e = V_e$ after normalization.
- **Observable effort refinement.** Observing effort improves contracting because the principal can separate average effort incentives from state-contingent adjustment incentives. But even then, first-best requires perfect correlation between P_e and V_e .
- **Applications.** The model explains why employees game performance measures, when revenue commissions work or fail, and when relative performance evaluation distorts incentives.
- **Economic takeaway.** Incentive design is not only about risk sharing. It is also about whether a performance measure gives the agent the right state-contingent instructions.

2 Incremental contribution of the paper

Baker's incremental contribution is to make **performance measurement** itself the central object of incentive theory. Instead of asking whether additional signals should supplement a contractible objective, the paper asks what happens when the principal's true objective cannot be used at all. The paper gives a simple formal metric for performance-measure quality: compare the marginal product of effort on the measure, P_e , to the marginal product of effort on value, V_e .

2.1 Contribution relative to Holmstrom's informativeness principle

- Holmstrom's informativeness principle asks whether an additional signal improves a contract that already uses the principal's objective.
- Baker studies the more basic problem where the principal's objective is not contractible, so the contract must be based on a proxy.
- The paper therefore shifts attention from signal informativeness about effort to **alignment of marginal incentives**.

- A performance measure can be easy to observe and highly precise, yet still be bad if its marginal response to effort differs from the true value response.

Interpretation: The paper’s central insight is that a performance measure is not evaluated by whether it measures activity accurately, but by whether it directs the agent toward actions the principal would choose if the principal had the agent’s information.

2.2 Contribution relative to risk-sharing models of incentive pay

- In standard principal-agent models, incentive intensity is often dampened because agents are risk averse.
- Baker shows that incentive intensity can be dampened even with risk neutrality.
- The reason is not income risk but **effort-distortion risk**: high-powered incentives on a bad proxy make the agent work hard in the wrong states and not hard enough in the right states.
- Convex effort costs make these mistakes costly, so the principal optimally lowers the piece rate.

Interpretation: The paper separates incentive distortion from risk sharing. Low-powered incentives may be optimal because high-powered incentives would cause the agent to misuse private information.

2.3 Contribution to the theory of gaming

- The paper formalizes the idea that employees can game performance measures.
- Gaming is not merely unethical behavior or manipulation. It is the rational response of an informed agent to incentives based on an imperfect proxy.
- When the agent knows which actions raise the proxy without raising value, high-powered incentives lead to dysfunctional effort allocation.
- The optimal contract weakens incentives to limit gaming.

Interpretation: The model explains why many organizations deliberately use weaker incentives than simple productivity logic would suggest.

2.4 Contribution to revenue commissions and performance-measure difficulty

- The paper clarifies what it means for performance to be difficult to measure.
- The problem is not that revenue is unavailable or expensive to measure. Revenue is usually easy to measure.
- The issue is whether revenue’s marginal response to salesperson effort matches the firm’s true marginal value.
- Revenue commissions work well for standardized products with cost unaffected by salesperson choices.
- They work poorly when salesperson choices affect both revenue and cost, quality, customer mix, or long-term value.

Interpretation: The paper turns the vague idea of “metering difficulty” into a precise marginal-alignment condition.

2.5 Contribution to relative performance evaluation

- The paper refines the theory of relative performance evaluation.
- Relative performance evaluation is nondistorting under unobservable effort only if the worker's effort does not affect the reference group.
- With observable effort, a relative measure can be nondistorting under weaker conditions: P_e and V_e need only be perfectly correlated.
- This distinction clarifies why relative performance can be useful in some settings but can induce sabotage or distorted effort in others.

Interpretation: The contribution is to distinguish variance reduction from incentive alignment. Relative performance can reduce noise but still distort actions if the reference group is affected by the agent.

3 Model objects and measurement

3.1 Principal's objective

The principal's true objective is

$$V(e, \epsilon),$$

where e is the agent's action or effort and ϵ is a vector of state variables.

- V is value before the agent's compensation.
- V is not contractible in the main model.
- $V_e \equiv \partial V(e, \epsilon) / \partial e$ is the marginal product of effort on value.
- Since ϵ is random from the principal's ex ante perspective, V_e is also a random variable.

Interpretation: The principal would like the agent to choose effort based on V_e , but the contract cannot be written directly on V or on V_e .

3.2 Performance measure

The contractible performance measure is

$$P(e, \epsilon),$$

with marginal product

$$P_e \equiv \frac{\partial P(e, \epsilon)}{\partial e}.$$

- P may be patents, revenue, accounting profit, divisional output, relative performance, or target achievement.
- P is contractible, but it is not the principal's true objective.
- P_e is the state-contingent marginal incentive signal sent to the agent.

Interpretation: The key question is whether P_e points in the same direction as V_e in each state of the world.

3.3 Information structure

- Neither principal nor agent observes ϵ before the contract is signed.
- After signing, but before effort choice, the agent observes ϵ .
- The principal cannot renegotiate and cannot observe enough to directly tell the agent the state-contingent optimal action.
- The agent uses the contract's piece rate and the realized P_e to decide how much effort to exert.

Interpretation: The performance measure is the principal's indirect communication device. It tells an informed agent how to use her private information.

3.4 Linear contract

The contract is

$$\text{agent's payoff} = S + bP(e, \epsilon),$$

where S is fixed salary and b is the piece rate.

- Linear contracts make incentive intensity summarized by a single parameter.
- The model focuses on how the optimal b depends on the relationship between P_e and V_e .
- The performance measure is normalized so that the average marginal product of effort on P equals that on V .

4 Models and theoretical specifications

4.1 Normalization of the performance measure

The paper scales the performance measure so that

$$\mathbb{E}[P_e(e, \epsilon)] = \mathbb{E}[V_e(e, \epsilon)]. \quad (1)$$

Interpretation: This normalization makes the piece rate interpretable. A piece rate of one corresponds to first-best incentives if P responds to effort in the same way as V in every state.

4.2 Agent utility and incentive compatibility

The risk-neutral agent's utility is

$$U = S + bP(e, \epsilon) - C(e), \quad C' > 0, \quad C'' > 0.$$

The participation constraint is

$$H \leq \mathbb{E}[S + bP - C(e)]. \quad (2)$$

The agent chooses effort after observing the state, so the incentive constraint is

$$bP_e(e^*, \epsilon) = C'(e^*). \quad (3)$$

Interpretation: The agent's effort choice is state-contingent because P_e is state-contingent. The principal does not directly know whether the agent works too much or too little in each state.

4.3 Principal's program

The principal chooses S and b to maximize

$$\max_{S,b} \mathbb{E}[V(e^*, \epsilon) - S - bP(e^*, \epsilon)] \quad (4)$$

subject to participation and incentive compatibility.

Interpretation: Salary adjusts to satisfy participation, while the piece rate trades off effort provision against incentive distortion.

4.4 Unobservable effort: optimal piece rate

The first-order solution can be expressed as

$$b^* = \frac{\mathbb{E}[V_e e_b^*]}{\mathbb{E}[P_e e_b^*]}, \quad (5)$$

where e_b^* is the derivative of the agent's effort choice with respect to the piece rate.

Differentiating the incentive condition gives

$$e_b^* = \frac{P_e}{C'' - bP_{ee}}. \quad (6)$$

With the paper's second-order approximation,

$$b^* = \frac{\mathbb{E}[V_e P_e]}{\mathbb{E}[P_e^2]}. \quad (7)$$

Under the normalization $\mathbb{E}[V_e] = \mathbb{E}[P_e] = 1$,

$$b^* = \frac{\text{Cov}(V_e, P_e) + 1}{\text{Var}(P_e) + 1} = \frac{\rho \sigma_{V_e} \sigma_{P_e} + 1}{\sigma_{P_e}^2 + 1}. \quad (8)$$

Interpretation: The optimal piece rate is a regression-like slope of true marginal value on measured marginal performance. The higher the correlation between V_e and P_e , the higher the optimal incentive intensity.

4.5 First-best condition

With unobservable effort, first-best requires

$$P_e(\epsilon) = V_e(\epsilon) \quad \text{for every state } \epsilon.$$

- Perfect correlation alone is not enough if variances differ under the unobservable-effort contract.
- The performance measure must have the same state-contingent marginal effect as value.
- The levels of P and V do not have to be identical; only their marginal responses to effort must match.

Interpretation: This is the paper's sharpest result. Incentives are efficient only if the performance measure gives exactly the same state-contingent instructions as the principal's objective.

4.6 Observable effort and the two-piece-rate contract

When effort is observable, the principal can use

$$\text{payoff} = S + b_1 \mathbb{E}[P \mid e] + b_2 (P - \mathbb{E}[P \mid e]). \quad (1)$$

The optimal piece rates are

$$b_1^* = \frac{\mathbb{E}[V_e]}{\mathbb{E}[P_e]} = 1, \quad (9)$$

$$b_2^* = \rho \frac{\sigma_{V_e}}{\sigma_{P_e}}. \quad (10)$$

Interpretation: Observable effort allows the contract to separate average effort provision from state-contingent information use. The first term induces the right average effort; the second term tells the agent how to adjust effort after observing the state.

4.7 Why even negative correlation can yield a positive piece rate

Equation (8) can yield $b^* > 0$ even if $\rho < 0$.

- The piece rate has two roles: providing average effort and using the agent’s information.
- Since effort is on average valuable, the principal may still want a positive piece rate.
- But bad alignment reduces the piece rate below one.

Interpretation: The model is not simply “pay only if the proxy is positively correlated.” It is about balancing average effort provision against distorted state-contingent effort allocation.

5 Theoretical design and identification logic

5.1 What identifies the mechanism in the model

- The mechanism is identified theoretically by comparing V_e and P_e .
- V_e is the state-contingent marginal value of effort to the principal.
- P_e is the state-contingent marginal reward signal faced by the agent.
- Distortion arises when these two marginal products diverge.

Interpretation: The model’s “identification” is mathematical rather than empirical. It isolates the performance-measure problem by assuming risk neutrality and showing that distortions remain.

5.2 Why asymmetric information matters

- If the agent did not learn useful state information before acting, the principal could prescribe or induce the correct average effort more easily.
- If the principal knew the state, the principal could design state-contingent instructions or adjust pay.
- The key tension is that the agent has useful private information but the contractible performance measure may direct that information toward the wrong action.

Interpretation: The performance measure is valuable because the agent has information. The same information makes misaligned measures dangerous because the agent knows how to exploit them.

5.3 Why risk neutrality is important

- By assuming risk neutrality, Baker removes the usual risk-sharing explanation for weak incentives.
- If the piece rate is below one, it is not because the agent dislikes income risk.
- The reduction in incentives is caused by incentive distortion and convex effort costs.

Interpretation: This makes the paper’s result stronger: imperfect performance measurement alone can explain low-powered incentives.

5.4 What the model cannot fully prove

- The paper is theoretical and does not estimate ρ , σ_{V_e} , or σ_{P_e} empirically.
- It assumes linear contracts and a single effort dimension for tractability.
- It does not model dynamic learning, renegotiation, or relational contracting.
- It abstracts from risk aversion, wealth constraints, and multitasking in the main formal model, although the intuition extends naturally to those settings.

Interpretation: The paper’s contribution is a clean metric and intuition, not a complete empirical theory of compensation design.

6 Section-by-section study guide

6.1 Section I: Introduction

- Read this section as the motivation for why performance measurement is central.
- The key contrast is between contractible objectives and imperfect proxies.
- The introduction also previews the main applications: gaming, revenue commissions, and relative performance evaluation.

6.2 Section II: Performance measurement and asymmetric information

- This section defines $V(e, \epsilon)$ and $P(e, \epsilon)$.
- It introduces the asymmetric information assumption: the agent observes ϵ before choosing effort.
- It emphasizes that marginal products V_e and P_e are random variables.

6.3 Section III: Derivation of the optimal contract

- This is the core theoretical section.
- Unobservable effort yields the regression-like piece rate in equation (8).
- Observable effort yields a two-piece-rate contract with $b_1 = 1$ and $b_2 = \rho\sigma_{V_e}/\sigma_{P_e}$.
- The main lesson is that ρ is a metric for performance-measure quality.

6.4 Section IV: Applications

- The patent example explains gaming.
- The sales example explains when revenue commissions are good or bad.
- The relative-performance example explains when peer comparisons distort incentives.

6.5 Section V and Appendix

- The conclusion summarizes the compensation-system implications.
- The appendix proves the observable-effort first-best result under perfect correlation.

7 Figures, tables, and theoretical result blocks

The paper has no empirical figures or tables. Therefore, this visual/results section follows the same notes format but uses the paper's main theoretical result blocks instead of extracted table or figure panels.

7.1 Result Block 1: Normalization and performance-measure quality

Result Block 1: Normalization of the performance measure

$$\mathbb{E}[P_e] = \mathbb{E}[V_e].$$

Read:

- The performance measure is scaled so that its average marginal product equals the average marginal product of true value.
- This makes the first-best piece rate equal to one in the ideal case.
- The normalization separates scale from alignment.

Detailed interpretation: The issue is not whether patents, revenue, or accounting profit have the right units. The issue is whether, after rescaling, the measure tells the agent to work more in exactly the states where effort creates more true value. A badly aligned measure can look reasonable on average while still producing wrong state-by-state incentives.

Economic message: A performance measure can be unbiased on average and still be a bad incentive instrument.

7.2 Result Block 2: Unobservable effort and the optimal piece rate

Result Block 2: Optimal piece rate with unobservable effort

$$b^* = \frac{\mathbb{E}[V_e P_e]}{\mathbb{E}[P_e^2]} = \frac{\text{Cov}(V_e, P_e) + 1}{\text{Var}(P_e) + 1}.$$

Read:

- b^* rises when P_e and V_e are more aligned.
- b^* falls when P_e is highly variable in ways unrelated to V_e .
- A measure with high variance can be dangerous if the variance reflects gaming opportunities rather than true value creation.

Detailed interpretation: The formula says the principal uses a performance measure as if it were a noisy state-contingent forecast of true marginal value. If the proxy's marginal effect is high when true marginal value is high, the principal can safely provide strong incentives. If the proxy's marginal effect is high when true marginal value is low, high-powered incentives induce misallocation of effort.

Economic message: Incentive intensity is a function of marginal alignment, not simply observability or measurement precision.

7.3 Result Block 3: First-best condition

Result Block 3: First-best condition under unobservable effort

$$P_e(\epsilon) = V_e(\epsilon) \quad \text{for all states } \epsilon.$$

Read:

- First-best requires identical state-contingent marginal incentives.
- It is not enough for P and V to be correlated in levels.
- It is not enough for the measure to be correct on average.
- The marginal response to effort must match state by state.

Detailed interpretation: This condition is stricter than many practical compensation schemes satisfy. A division manager's accounting profit may be highly correlated with firm value, but if the manager can raise accounting profit by cutting maintenance or quality, P_e differs from V_e in exactly the states where incentives matter.

Economic message: The model explains why contracts based on seemingly sensible proxies can still create costly distortions.

7.4 Result Block 4: Observable effort and two-piece-rate contracts

Result Block 4: Observable effort contract

$$\begin{aligned} \text{Payoff} &= S + b_1 \mathbb{E}[P \mid e] + b_2 (P - \mathbb{E}[P \mid e]), \\ b_1^* &= 1, \quad b_2^* = \rho \frac{\sigma_{V_e}}{\sigma_{P_e}}. \end{aligned}$$

Read:

- Observing effort lets the principal pay separately for average effort and state-contingent performance deviations.
- b_1 gives the correct average effort incentives.
- b_2 controls how much the agent should respond to the state-specific performance signal.
- If $\rho = 1$, the state-specific signal is perfectly aligned with true marginal value.

Detailed interpretation: Observable effort does not automatically solve the agency problem because the principal still does not know the state-contingent value of effort. But it improves the contract because the principal can mean-adjust the performance measure. The incentive on the residual part is exactly a regression coefficient of V_e on P_e .

Economic message: Monitoring effort helps, but it does not remove the need for a good performance measure.

7.5 Result Block 5: Gaming performance measures

Result Block 5: Patent example

The optimal patent piece rate is below the average value of a patent when patent difficulty varies and the scientist knows which projects are easy.

Read:

- The firm values innovation but cannot contract directly on true research value.
- It uses patents as a proxy.
- The scientist knows which patent opportunities are easy or difficult.
- A full-value patent bonus induces too much effort on easy low-value patenting opportunities and too little effort on harder high-value ones.

Detailed interpretation: Gaming occurs because the proxy has a state-contingent marginal product that differs from true value. The agent is not necessarily shirking; she is optimizing the contract. The firm weakens incentives because stronger incentives would increase the agent's tendency to chase the proxy rather than value.

Economic message: High-powered incentives are dangerous when the agent has better information about how to manipulate the measure.

7.6 Result Block 6: Revenue-based commissions

Result Block 6: Revenue commissions

Revenue commissions are first-best for a standardized product if salesperson effort affects revenue but not cost. They are distorted for complex products if effort also affects cost, quality, or product configuration.

Read:

- Firm A sells a standardized product; costs are independent of salesperson effort.
- Revenue is then a good proxy because marginal revenue and marginal value move together.
- Firm B sells a complex product; salesperson effort can affect both quantity and cost.
- Revenue alone then misses the cost side and can induce value-destroying sales.

Detailed interpretation: The same measure, revenue, can be good in one environment and bad in another. What matters is whether incremental revenue equals incremental value. If salespeople can sell expensive-to-serve customers or customize products inefficiently, revenue commissions become distorted.

Economic message: Performance-measure difficulty means lack of marginal alignment, not lack of observability.

7.7 Result Block 7: Relative performance evaluation

Result Block 7: Relative performance

$$P(e, \epsilon) = V(e, \epsilon) - R(e, \epsilon).$$

Unobservable effort is nondistorting only when $R_e = 0$. Observable effort can be nondistorting when P_e and V_e are perfectly correlated.

Read:

- Relative performance subtracts reference-group output from individual output.
- It can reduce common noise in performance.
- It distorts incentives if the worker can affect the reference group.
- With observable effort, the condition becomes less strict because the contract can separate average effort from residual performance deviations.

Detailed interpretation: Relative performance evaluation is useful for risk reduction, but it can create sabotage or other strategic actions toward peers. Baker's framework clarifies when the distortion is severe and when the risk-reduction benefit may justify a small distortion.

Economic message: Peer benchmarking is valuable only when it filters noise without changing the agent's desired marginal actions.

8 Detailed result map

- **Performance-measure normalization:** scale P so a first-best piece rate is one when $P_e = V_e$.
- **Unobservable effort:** optimal incentives are dampened according to the covariance between P_e and V_e .
- **First-best benchmark:** first-best requires exact marginal equality, not merely level correlation.
- **Observable effort:** a richer contract can use effort and deviations from expected performance.
- **Gaming:** agents exploit private information about when the proxy is easy to improve.

- **Revenue commissions:** revenue is efficient only when marginal revenue equals marginal value.
- **Relative evaluation:** reference groups reduce noise but can distort incentives when the agent affects peers.

9 Short summary

- Baker studies incentive contracts when the principal’s true objective is not contractible.
- The principal uses a contractible proxy $P(e, \epsilon)$.
- The agent observes the state ϵ before choosing effort.
- The agent responds to P_e , not directly to V_e .
- Incentives are efficient only if the performance measure’s marginal response to effort matches true value’s marginal response.
- The optimal piece rate is lower when the performance measure is poorly aligned with value.
- This dampening occurs even if the agent is risk neutral.
- The paper provides a correlation-based metric for performance-measure quality.
- The framework explains gaming, weak incentives, revenue-commission problems, and relative-performance distortions.
- The main lesson is that incentive pay must be designed around marginal alignment, not just observability or average value.

10 Conclusion

10.1 Core conclusion

Baker’s core conclusion is that imperfect performance measurement can distort incentives even without risk aversion. If a contract is based on a proxy rather than the principal’s true objective, high-powered incentives can cause the agent to use private information in ways that improve the proxy but reduce value. The optimal contract therefore often dampens incentives.

10.2 Economic intuition

The agent observes information about the state of the world before acting. This information is valuable because it allows the agent to adjust effort. But the agent adjusts effort according to the performance measure, not according to the principal’s objective. If the measure and objective have different marginal responses to effort, the agent’s private information becomes a source of distortion.

10.3 Why the paper matters

The paper explains why organizations frequently use weak incentives, salaries, monitoring, partnerships, stock ownership, or richer performance systems instead of simple high-powered bonus schemes. Weak incentives may be efficient when available measures are easy to manipulate or only loosely connected to value.

10.4 Implications for compensation design

- Use high-powered incentives when the performance measure's marginal response to effort is closely aligned with value.
- Use salary, monitoring, or richer qualitative evaluation when available measures are poorly aligned.
- Use multiple measures when each captures a different margin of value creation.
- Avoid simple commissions when agents can raise measured output by imposing hidden costs.
- Treat gaming as a design problem, not merely a behavioral anomaly.

10.5 Limitations

The model is intentionally simple. It uses linear contracts, risk neutrality, and a single effort dimension. It does not estimate empirical correlations between P_e and V_e , model repeated relationships, or incorporate renegotiation. These simplifications make the metric clear but leave open how organizations should measure marginal alignment in practice.

10.6 Takeaway

The most important lesson is that a performance measure is good only if it gives the agent the right marginal instructions. A proxy can be accurate on average, cheap to observe, and highly correlated with value in levels, but still generate bad incentives. The design of compensation systems should therefore start with the question: *When this measure rises because the agent changes behavior, does true value rise by the same amount?*